

JINWOO KIM

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Research Interest

My focus is in making deep learning models generalize better beyond their training data so that they can be used to solve challenging problems, such as those in scientific domains. I am studying this problem from the complementary perspectives of invariance and equivariance, parallel neural processing, and Solomonoff induction. I often use tools from theories of graphs, manifolds, (semi)groups and categories, as well as Markov processes such as random walks, diffusions, and flows. Recently, I am focused on flow language models, latent reasoning, synthetic data, and vision pretraining.

Academic Placement and Employment

KAIST , Postdoctoral Researcher	03/2026 – 02/2027
New York University , Visiting Scholar (host: Kyunghyun Cho, Rajesh Ranganath)	11/2025 – 01/2026
LG AI Research , Research Intern (host: Moontae Lee, Honglak Lee)	01/2022 – 07/2022

Education

KAIST , M.S./Ph.D. in Computer Science (advisor: Seunghoon Hong) - Ph.D. Thesis: Architecture-Agnostic Invariances for Deep Learning KAIST CoE Best Ph.D. Dissertation Award	03/2021 – 02/2026
KAIST , B.S. in Brain Engineering and Computer Science (double major) - GPA 4.05/4.3 (Summa Cum Laude)	03/2016 – 02/2021

Publications

C: conference, J: journal, W: workshop, P: preprint, *: equal contribution, †: equal advising

P1. Thinking with Looped Flows

Ayhan Suleymanzade, Chanhyuk Lee, Floor Eijkelboom, Nicholas M. Boffi, İsmail İlkan Ceylan[†], Jinwoo Kim[†]
arXiv 2026

C17. MUX: Continuous Reasoning via Multiplexed Tokens

Ayhan Suleymanzade, Halil Alperen Gozeten, Michael Bronstein, İsmail İlkan Ceylan[†], Jinwoo Kim[†]
NeurIPS 2026 **Spotlight (292/30709=0.95%)**

C16. Flow Map Language Models: One-step Language Modeling via Continuous Denoising

Chanhyuk Lee, Jaehoon Yoo, Manan Agarwal, Sheel Shah, Jerry Huang, Aditi Raghunathan, Seunghoon Hong, Nicholas M. Boffi[†], Jinwoo Kim[†]
NeurIPS 2026 / COLM 2026 NonAR-LM Workshop **Spotlight**

C15. Multi-View Relational Distillation for Spatial Reasoning with Vision-Language Models

Kiet T. Nguyen, Hanbo Shim, Jinwoo Kim[†], Seunghoon Hong[†]
NeurIPS 2026

C14. RelAgent: LLM Agents as Data Scientists for Relational Learning

Xingyue Huang, Louis Tichelman, Jinwoo Kim, Krzysztof Olejniczak, İsmail İlkan Ceylan
NeurIPS 2026

W3. Self-conditioned Flow Map Language Models via Fixed-point Flows

Jaehoon Yoo*, Wonjung Kim*, Floor Eijkelboom, Chanhyuk Lee, Nicholas M. Boffi, Seunghoon Hong[†], Jinwoo Kim[†]
COLM 2026 NonAR-LM Workshop **Oral**

W2. Posterior Refinement: Fast Language Generation via Any-Order Flow Maps

Manan Agarwal*, Sheel Shah*, Chanhyuk Lee, Jaehoon Yoo, Jerry Huang, Seunghoon Hong, Aditi Raghunathan, Jinwoo Kim[†], Nicholas M. Boffi[†]
COLM 2026 NonAR-LM Workshop **Spotlight**

- W2. **What are the Right Symmetries for Formal Theorem Proving?**
Krzysztof Olejniczak, Radoslav Dimitrov, Xingyue Huang, Bernardo Cuenca Grau, Jinwoo Kim[†], İsmail İlkan Ceylan[†]
ICML 2026 AI4Math Workshop
- C13. **Inverting Data Transformations via Diffusion Sampling**
Jinwoo Kim^{*}, Sékou-Oumar Kaba^{*}, Jiyun Park, Seunghoon Hong[†], Siamak Ravanbakhsh[†]
ICML 2026
- C12. **Flock: A Knowledge Graph Foundation Model via Learning on Random Walks**
Jinwoo Kim^{*}, Xingyue Huang^{*}, Krzysztof Olejniczak, Kyunghin Min, Michael Bronstein, Seunghoon Hong, İsmail İlkan Ceylan
ICLR 2026
- C11. **Sequence Modeling with Spectral Mean Flows**
Jinwoo Kim, Max Beier, Petar Bevanda, Nayun Kim, Seunghoon Hong
NeurIPS 2025
- C10. **Revisiting Random Walks for Learning on Graphs**
Jinwoo Kim, Olga Zaghen^{*}, Ayhan Suleymanzade^{*}, Youngmin Ryou, Seunghoon Hong
ICLR 2025 **Spotlight (380/11672=3.26%)** / **ELLIS Mobility Grant, ICML 2024 GRaM Workshop**
- C9. **3D Denoisers are Good 2D Teachers: Molecular Pretraining via Denoising and Cross-Modal Distillation**
Sungjun Cho, Dae-Woong Jeong, Sung Moon Ko, Jinwoo Kim, Sehui Han, Seunghoon Hong, Honglak Lee, Moontae Lee
AAAI 2025 **Oral**
- C8. **Simulation-Free Training of Neural ODEs on Paired Data**
Semin Kim^{*}, Jaehoon Yoo^{*}, Jinwoo Kim, Yeonwoo Cha, Saehoon Kim, Seunghoon Hong
NeurIPS 2024
- W1. **Learning Symmetrization for Equivariance with Orbit Distance Minimization**
Tien Dat Nguyen^{*}, Jinwoo Kim^{*}, Hongseok Yang, Seunghoon Hong
NeurIPS 2023 NeurReps Workshop
- C7. **Learning Probabilistic Symmetrization for Architecture Agnostic Equivariance**
Jinwoo Kim, Tien Dat Nguyen, Ayhan Suleymanzade, Hyeokjun An, Seunghoon Hong
NeurIPS 2023 **Spotlight (378/12345=3.06%)**
- C6. **Universal Few-shot Learning of Dense Prediction Tasks with Visual Token Matching**
Donggyun Kim, Jinwoo Kim, Seongwoong Cho, Chong Luo, Seunghoon Hong
ICLR 2023 **Outstanding Paper Award (4/4955=0.08%)** / **Silver Prize, Samsung Humantech Paper Award, 2023**
- C5. **Pure Transformers are Powerful Graph Learners**
Jinwoo Kim, Tien Dat Nguyen, Seonwoo Min, Sungjun Cho, Moontae Lee, Honglak Lee[†], Seunghoon Hong[†]
NeurIPS 2022
- C4. **Transformers meet Stochastic Block Models: Attention with Data-Adaptive Sparsity and Cost**
Sungjun Cho, Seonwoo Min, Jinwoo Kim, Moontae Lee, Honglak Lee, Seunghoon Hong
NeurIPS 2022
- C3. **Equivariant Hypergraph Neural Networks**
Jinwoo Kim, Saeyoon Oh, Sungjun Cho, Seunghoon Hong
ECCV 2022
- C2. **Transformers Generalize DeepSets and Can be Extended to Graphs and Hypergraphs**
Jinwoo Kim, Saeyoon Oh, Seunghoon Hong
NeurIPS 2021 / **Qualcomm Innovation Fellowship Korea, 2021**
- C1. **SetVAE: Learning Hierarchical Composition for Generative Modeling of Set-Structured Data**
Jinwoo Kim^{*}, Jaehoon Yoo^{*}, Juho Lee, Seunghoon Hong
CVPR 2021
- J1. **Spontaneous Retinal Waves Can Generate Long-Range Horizontal Connectivity in Visual Cortex**
Jinwoo Kim^{*}, Min Song^{*}, Jaeson Jang, Se-Bum Paik
The Journal of Neuroscience 40(34) 2020

Honors

Awards

Best Ph.D. Dissertation Award, KAIST College of Engineering	2026
Outstanding Researcher Award, KAIST-Mila Prefrontal AI Research Center	2024
ELLIS Mobility Grant [C10], ICML 2024 GRaM Workshop	2024
Outstanding Paper Award [C6], ICLR 2023	2023
Silver Prize, Samsung Humantech Paper Award [C6]	2023
Qualcomm Innovation Fellowship Korea [C2]	2022
KAIST Engineering Innovator Award (five recipients), KAIST College of Engineering	2020

Scholarships and Fellowships

Kwanjeong Education Foundation Scholarship	2022 – 2023
Korea National Science & Technology Scholarship	2018 – 2019
KAIST Alumni Fellowship	2017 – 2020
KAIST Presidential Fellowship	2016 – 2020
Hansung Scholarship for Gifted Students	2015 – 2016

Invited Talks

Flow Language Models [C11, C16, W2, W3, P1]	
• MIT CSAIL (host: Hyunwoo Ryu)	10/2026
• POSTECH IME (host: Sangwoo Mo)	10/2026
• SNU IPAI Seminar (host: Jaesik Park, Pierre Musacchio)	07/2026
• Discrete Diffusion Reading Group (host: Subham Sahoo, Zhihan Yang)	04/2026
• Ben-Gurion University of the Negev (host: Ilan Naiman)	12/2025
Architecture-Agnostic Invariances for Deep Learning [C7, W1, C10, C12, C13]	
• Mila - Quebec AI Institute (host: Minsu Kim, Junyeob Baek)	07/2025
• KAIST AI899 Geometric DL (host: Sungsoo Ahn)	05/2025
• Mila - Quebec AI Institute (host: Siamak Ravanbakhsh, Sékou-Oumar Kaba)	12/2024
• KAIST-Mila Prefrontal AI Research Center (host: Sungjin Ahn)	11/2024
• Sungkyunkwan University (SKKU) (host: Chang Woo Myung)	08/2024
• Pohang University of Science and Technology (POSTECH) (host: Sungsoo Ahn)	11/2023
Visual Token Matching [C6]	
• KAIST–Samsung Electronics DS Division Exchange Meetup (host: Chulmoo Kang)	08/2023
Transformers for Sets, Graphs, and Hypergraphs [C1, C2, C5]	
• Microsoft USA (host: Nabiha Asghar)	01/2023
• Qualcomm Korea (host: Jaewon Choi)	01/2023
• NeurIPS 2022 at KAIST (host: Dongkwan Kim)	11/2022
• Learning on Graphs and Geometry Reading Group (LoGaG) (host: Hannes Stärk)	08/2022
• KAIST AI Workshop 21/22 (host: Dongkwan Kim)	01/2022
• NeurIPS 2021 Social: ML in Korea (host: Jung-Woo Ha)	12/2021
• Naver AI Author Meetup for CVPR 2021 (host: Jung-Woo Ha)	09/2021
• Korean Conference on Computer Vision 2021 (host: Jongwoo Lim)	09/2021
Retinal Waves and Prenatal Wiring of Primary Visual Cortex [J1]	
• Society for Neuroscience, Chicago, IL, US	10/2019

Academic Services

Area Chair: ICLR 2027
Workshop Area Chair: ICML 2026 GFM
Conference Reviewer: NeurIPS, ICLR, ICML, CVPR, ICCV, AISTATS, LoG, TAG-DS, IJCNN, ACCV
Journal Reviewer: TMLR, Neural Networks, IJCV

Teaching

Teaching Assistant, KAIST School of Computing

Undergraduate Research Program (URP)

2022, 2024

Introduction to Deep Learning (CS492I)

2021, 2022, 2023

Computer Vision (CS576)

2022, 2023

School of Computing Colloquium (CS966, CS986)

2021

Teaching Assistant, Samsung Electronics

Samsung Research AI Expert Program

2021 – 2024

Mentoring

Chanwoo Kim, M.S. student @ KAIST

2026

Toghrul Abbasov, B.S. student @ KAIST

2026

Kiet T. Nguyen, Ph.D. student @ KAIST [C14]

2026

Chanhyuk Lee, M.S. student @ KAIST [C16, W2, W3, P1]

2025 – 2026

Kyungbin Min, B.S. student @ KAIST [C12] → M.S. student @ KAIST

2025 – 2026

Jiyeon Park, B.S. student @ KAIST [C13] → M.S. student @ KAIST

2024 – 2026

Nayun Kim, B.S. student @ KAIST [C11] → M.S. student @ KAIST

2024 – 2026

Ayhan Suleymanzade, B.S. student @ KAIST [C7, C10] → Ph.D. student @ EPFL [C17, P1]

2023 – 2026

Youngmin Ryou, B.S. student @ KAIST [C10] → M.S. Student @ KAIST

2023 – 2026

Nicole Shen, B.S. student @ MIT → Intern @ MIT LIDS

2024

Semin Kim, M.S. student @ KAIST [C8] → Ph.D. student @ KAIST

2023 – 2024

Olga Zaghen, M.S. student @ UniTrento [C10] → Ph.D. student @ UvA Amsterdam

2023

Tien Dat Nguyen, B.S. student @ KAIST [C5, C7, W1] → M.S. student @ UWaterloo

2021 – 2023

Daniel Sungho Jung, B.S. student @ Penn State → Ph.D. student @ SNU

2021

Saeyoon Oh, B.S. student @ KAIST [C2, C3] → Engineer @ FuriosaAI

2021

Projects

Korea National Research Foundation (NRF), Physical Reasoning in Language Models

2024 – 2025

Korea Ministry of Science and ICT, Visual Commonsense Reasoning

2021 – 2023

Korea National Research Foundation (NRF), Cooperative Robotic Intelligence

2021 – 2023

References

Seunghoon Hong, Associate Professor at KAIST

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İsmail İlkan Ceylan, Associate Professor at TU Wien

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Nicholas M. Boffi, Assistant Professor at Carnegie Mellon University

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Kyunghyun Cho, Full Professor at New York University

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Siamak Ravanbakhsh, Associate Professor at McGill University

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Moontae Lee, Head of Superintelligence Lab at LG AI Research

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